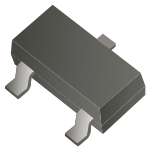
Two-Line ESD Protection

AGSOT24C-HF

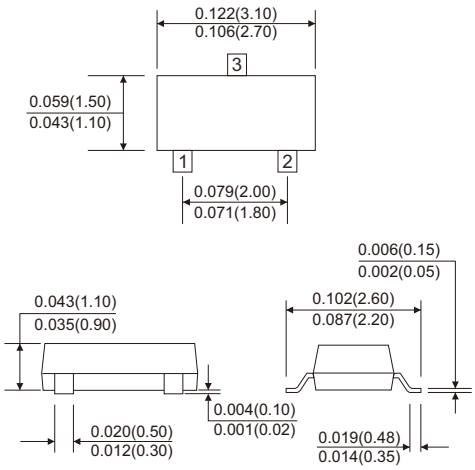
RoHS Device
Halogen Free



Features

- Two-line ESD-protection device.
- ESD-protection ace. IEC 61000-4-2.
  ±30 kV contact discharge
  ±30 kV air discharge
- AEC-Q101 Qualified.

SOT-23

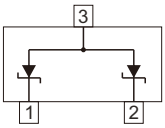


Dimensions in inches and (millimeter)

Mechanical data

- Case: SOT-23, molded plastic.

Circuit Diagram



Maximum Ratings (at T_A=25°C unless otherwise noted)

Parameter	Test Conditions	Symbol	Limits	Unit
Peak pulse power	Pin 1 to 3 or Pin 2 to 3 acc. IEC 61000-4-5, tp = 8/20 µs ; single shot	P_{pp}	235	W
	Pin 1 to 2 or Pin 2 to 1 ; Pin 3 not connected acc. IEC 61000-4-5, tp = 8/20 µs ; single shot		240	
Peak pulse current	Pin 1 to 3 or Pin 2 to 3 acc. IEC 61000-4-5, tp = 8/20 µs ; single shot	I_{pp}	5	A
	Pin 1 to 2 or Pin 2 to 1 ; Pin 3 not connected acc. IEC 61000-4-5, tp = 8/20 µs ; single shot		5	
ESD immunity	Contact discharge acc. IEC 61000-4-2 ; 10 pulses	V_{PP}	±30	kV
	Air discharge acc. IEC 61000-4-2 ; 10 pulses	V_{ESD}	±30	
Operating temperature range	Junction temperature	T_J	-55 ~ +150	°C
Storage temperature range		T_{STG}	-55 ~ +150	°C

Electrical Characteristics (at Ta=25°C unless otherwise noted)

Between Pin 1 to Pin 3 or Pin 2 to Pin 3

Parameter	Conditions	Symbol	Min	Typ	Max	Unit
Protection paths	Number of lines which can be protected	$N_{channel}$			2	lines
Reverse stand-off voltage	I_R = 1μA	V_{RWM}			24	V
Reverse current	V_R = 24V	I_R			1	μA
Reverse breakdown voltage	I_R = 1mA	V_{BR}	27	30		V
Clamping voltage	I_{PP} = 1A I_{PP} = I_{PPM} = 5A	V_C		34 41	41 47	V
Forward clamping voltage	I_{PP} = 1A I_{PP} = I_{PPM} = 5A	V_F		1 1.4	1.2 -	μA
Junction capacitance	V_R = 0V, f = 1MHz V_R = 12V, f = 1MHz	C_D		65 20	80 -	pF

Between Pin 1 to Pin 2 or Pin 2 to Pin 1 ; Pin 3 not Connected

Parameter	Conditions	Symbol	Min	Typ	Max	Unit
Protection paths	Number of lines which can be protected	$N_{channel}$			1	lines
Reverse stand-off voltage	I_R = 1μA	V_{RWM}			24.5	V
Reverse current	V_R = 24.5V	I_R			1	μA
Reverse breakdown voltage	I_R = 1mA	V_{BR}	27.5	30.7		V
Clamping voltage	I_{PP} = 1A I_{PP} = I_{PPM} = 5A	V_C		34 40	41 48	V
Junction capacitance	V_R = 0V, f = 1MHz V_R = 12V, f = 1MHz	C_D		33 18	40 -	pF

Rating and Characteristic Curves (AGSOT24C-HF)

Fig.1 - Typical Forward Current vs. Forward Voltage

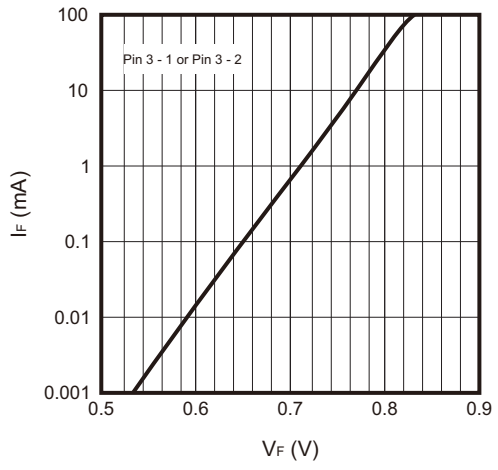
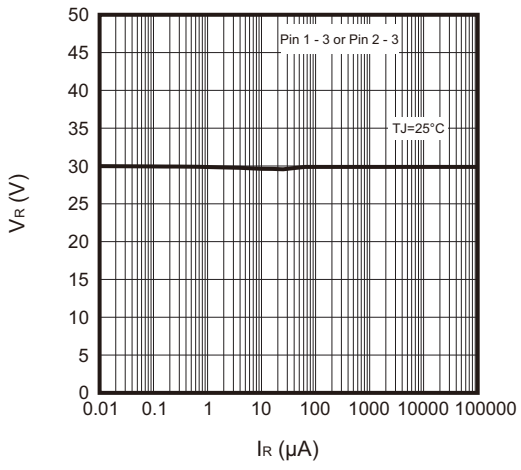
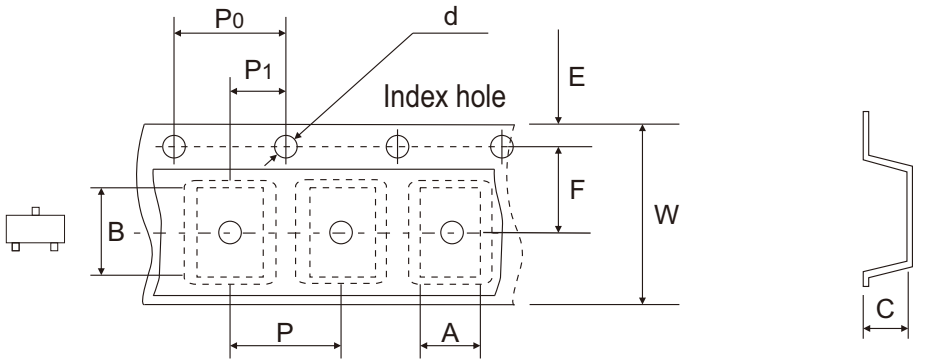
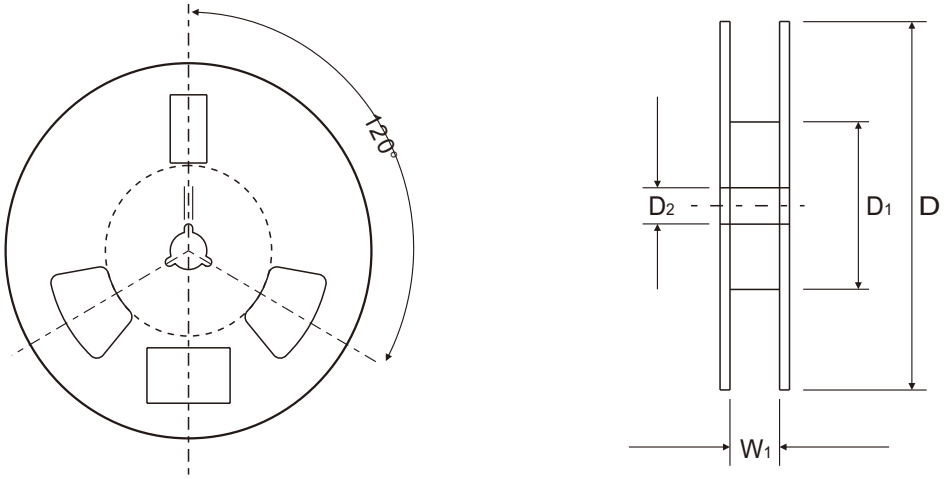


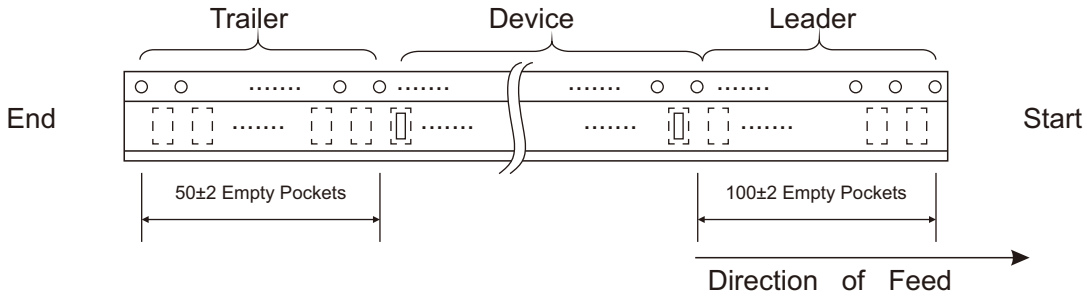
Fig.2 - Typical Reverse Voltage vs. Reverse Current



Reel Taping Specification







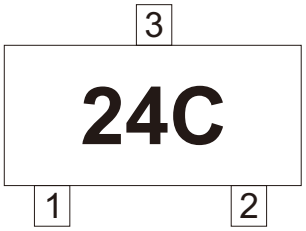
	SYMBOL	A	B	C	d	D	D1	D2
SOT-23	(mm)	3.15 ± 0.10	2.77 ± 0.10	1.22 ± 0.10	1.50 ± 0.10	178.00 ± 1.00	54.00 ± 0.50	13.00 ± 0.50
	(inch)	0.124 ± 0.004	0.109 ± 0.004	0.048 ± 0.004	0.059 ± 0.004	7.008 ± 0.039	2.126 ± 0.020	0.512 ± 0.020

	SYMBOL	E	F	P	P0	P1	W	W1
SOT-23	(mm)	1.75 ± 0.10	3.50 ± 0.05	4.00 ± 0.10	4.00 ± 0.10	2.00 ± 0.05	8.00 + 0.30 − 0.10	9.50 ± 1.00
	(inch)	0.069 ± 0.004	0.138 ± 0.002	0.157 ± 0.004	0.157 ± 0.004	0.079 ± 0.002	0.315 + 0.012 − 0.004	0.374 ± 0.039

Company reserves the right to improve product design , functions and reliability without notice.

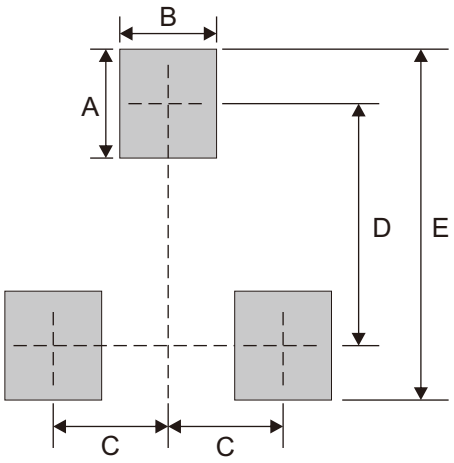
Marking Code

Part Number	Marking Code
AGSOT24C-HF	24C



Suggested PAD Layout

SIZE	SOT-23	
	(mm)	(inch)
A	0.90	0.035
B	0.80	0.031
C	0.95	0.037
D	2.00	0.079
E	2.90	0.114



Note: 1. The pad layout is for reference purposes only.

Standard Packaging

Case Type	REEL PACK	
	REEL (pcs)	Reel Size (inch)
SOT-23	3,000	7